

THE WHOLE OF ADVANCED HIGHER MATHS 2026 in 5 HOURS!

**FULL VIDEO on
YOUTUBE**

**DO QUESTIONS on THE
FREE APP**

*Copyright Notice : Exam Questions : Copyright ©
Qualifications Scotland
Solutions by clellandmaths and not produced or
endorsed by SQA*

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!



JOIN NOW

STOP PANICKING. START PREPARING.



JOIN THE AH MATHS LAST MINUTE LIVE STREAM.

Working through these SQA past papers is a great start. But if you want to know exactly what to expect on May 8th, you need to be in the last-minute live stream.

What You Get

- 🧠 Mr. Clelland's 100% Original Predicted Questions.
- 💬 Live, real-time Q&A to solve your exact problems.
- ⌚ Final exam technique and time-management hacks.

Note: The Live Stream is an independent Clelland Maths event and uses entirely original material. It does not contain SQA copyrighted questions.

LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

JOIN THE CHANNEL!

(a) Express $\frac{-2}{x(x+1)}$ in partial fractions. 2

(b) Use integration by parts to find $\int xe^{3x} dx$. 3

(c) Using your answers to (a) and (b), solve

$$\frac{dy}{dx} - \frac{2y}{x(x+1)} = \frac{x^3 e^{3x}}{(x+1)^2}. \quad 5$$

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

PARTIAL FRACTIONS

(a) Express $\frac{-2}{x(x+1)}$ in partial fractions.

2

(b) Use integration by parts to find $\int xe^{3x} dx$.

3

(c) Using your answers to (a) and (b), solve

$$\frac{dy}{dx} - \frac{2y}{x(x+1)} = \frac{x^3 e^{3x}}{(x+1)^2}.$$

5

Express $\frac{2x^2 - 18x + 4}{(x-1)(x-3)(x+5)}$ in partial fractions.

3

DIFFERENTIATION – PRODUCT RULE

Differentiate the following with respect to x :

(a) $y = \cot 3x$

2

(b) $f(x) = 5x(4x-7)^{\frac{1}{2}}$.

2

Given $y = 7x \tan 2x$, find $\frac{dy}{dx}$.

2

DIFFERENTIATION – QUOTIENT RULE

Given $y = \frac{\sin 7x}{1+x^2}$, find $\frac{dy}{dx}$.

2

Given $f(x) = \frac{2x^3 + x}{3+2x}$, find $f'(x)$. Simplify your answer.

3

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

IMPLICIT DIFFERENTIATION

A curve is defined by the equation $2y^2 + 4xe^{2y} = 3x$.

Find an expression for $\frac{dy}{dx}$ in terms of x and y . 3

Calculate the gradient of the tangent to the curve with equation $x^2y^2 - 2y = \sin 3x$ at the point $(0, 0)$. 3

LOGARITHMIC DIFFERENTIATION

A curve is defined by $y = x^{5x^2}$, where $x > 0$.

Find $\frac{dy}{dx}$ in terms of x . 5

A curve is defined by $y = x^{\cot x}$.

Use logarithmic differentiation to find $\frac{dy}{dx}$.
Write your answer in terms of x . 5

PARAMETRIC DIFFERENTIATION

A curve is defined on a suitable domain by the equations $x = t^2$ and $y = \tan t$.

Find in terms of t :

- (a) $\frac{dy}{dx}$ 2
- (b) $\frac{d^2y}{dx^2}$. 3

A curve defined parametrically has the following properties:

- $x = \tan^{-1} 2t$
- $\frac{dy}{dx} = 6t(1 + 4t^2)$
- $y = 5$ when $t = 1$.

Find y in terms of t . 4

DIFFERENTIATION – RELATED RATES OF CHANGE

The volume, $V \text{ cm}^3$, of water in a tank is given by

$$V = \frac{1}{5}h^3, \text{ where } h \text{ cm is the depth of water in the tank.}$$

Water is being piped into the tank at a rate of $6 \text{ cm}^3/\text{second}$.

Water is leaking from the bottom of the tank at a rate of $\frac{1}{10}\sqrt{h} \text{ cm}^3/\text{second}$.

Calculate the rate of change of the depth of water when $h = 400$.

4

A metal rod is heated such that its volume increases at a constant rate of 12 mm^3 per minute.

The volume of the rod is modelled, throughout the process, by $V = 5\pi r^3$, where r is measured in millimetres.

Find the rate at which r is increasing when $r = 10$.

4

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

INTEGRATION PARTIAL FRACTIONS

Find the exact value of $\int_0^3 \frac{4}{2x+1} dx$. 2

Use partial fractions to find $\int \frac{3x-7}{x^2-2x-15} dx$. 4

INTEGRATION BY SUBSTITUTION

Use the substitution $u = \tan 2x$ to evaluate $\int_0^{\frac{\pi}{8}} \frac{\sqrt{\tan 2x}}{\cos^2 2x} dx$. 4

Find $\int \frac{x^2}{x^3+10} dx$. 2

INTEGRATION BY PARTS

Use integration by parts to find $\int x^4 \ln x dx$, $x > 0$. 3

Use integration by parts to find

$$\int e^{2x} \sin 5x dx. \quad 5$$

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

DIFFERENTIAL EQUATIONS – SEPARATE VARIABLES

Solve the differential equation

$$\frac{dy}{dx} = \frac{y}{2x-1}, \quad x, y > 1,$$

given that $y = 12$ when $x = 5$. Express y in terms of x .

5

A storage tank contains a mixture of salt and water. An additional amount of salt and water pours in while, at the same time, some of the existing mixture pours out.

The process can be modelled by the differential equation

$$\frac{dW}{dt} = \frac{36-W}{120}, \quad W < 36$$

where W is the amount of salt in kilograms at time t minutes.

Initially, the storage tank contains 8 kilograms of salt.

(a) Express W in terms of t .

5

(b) Find the rate at which the amount of salt is increasing after 67 minutes.

2

As the process continues, the amount of salt approaches a limit L kilograms.

(c) Find the value of L , justifying your answer.

1

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

DIFFERENTIAL EQUATIONS – FIRST ORDER

Find the general solution of the differential equation

$$\frac{dy}{dx} - \frac{2}{x}y = x^2 \sec^2 3x.$$

Give your answer in the form $y = f(x)$.

5

(a) Differentiate $x \ln x - x$ with respect to x .

2

(b) Hence find the general solution of the differential equation

$$\frac{dy}{dx} + y \ln x = x^{-x}.$$

4

DIFFERENTIAL EQUATIONS – SECOND ORDER

Solve the differential equation

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 15x^2 - 31x + 40$$

given that $y = 4$ and $\frac{dy}{dx} = 13$ when $x = 0$.

9

Solve the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 8y = 0$$

given that $y = -2$ and $\frac{dy}{dx} = 22$ when $x = 0$.

5

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

APPLICATIONS OF CALCULUS

RECTILINEAR MOTION

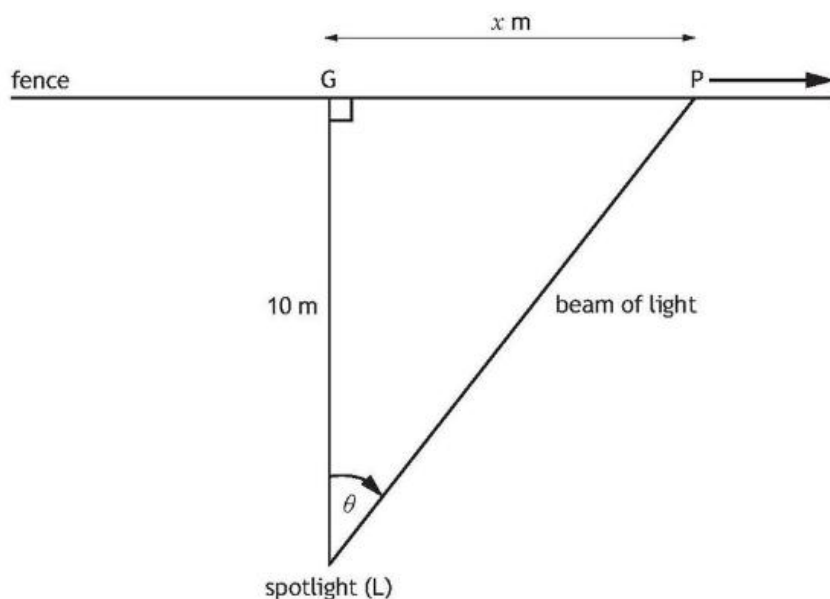
Relative to a fixed origin, the velocity, v metres per second, of an object at time t seconds is given by $v = 2t + e^{5t}$.

- (a) Find an expression for the displacement of the object, s metres, in terms of t , given that when $t = 0$, $s = 0$. 2
- (b) Show that the acceleration of the object is always positive. 2

A security spotlight is situated 10 metres from a straight fence. The spotlight rotates at a constant speed and makes one full revolution every 12 seconds.

The situation at time t seconds is modelled in the diagram below, where:

- L is the position of the spotlight
- G is the point on the fence nearest to the spotlight
- P is the position where the light hits the fence
- θ is the angle between LG and LP
- x is the distance in metres from G to P.



(a) Show that:

(i) $\frac{d\theta}{dt} = \frac{\pi}{6}$ radians per second 1

(ii) $\frac{dx}{dt} = \frac{5\pi}{3} \sec^2 \theta$ metres per second. 4

(b) Prove that $1 + \tan^2 \theta = \sec^2 \theta$. 1

(c) Hence, or otherwise, find the exact value of $\frac{dx}{dt}$ when P is 5 metres from G. 3

LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

JOIN THE CHANNEL!

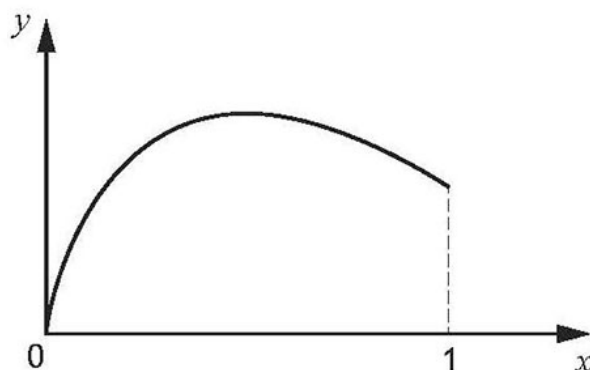
APPLICATIONS OF CALCULUS

VOLUME OF REVOLUTION

(a) Using the substitution $u = 2x^2$, or otherwise, find $\int xe^{-2x^2} dx$.

2

The diagram shows part of the curve with equation $y = \frac{4\sqrt{x}}{e^{x^2}}$.



A solid is generated by rotating the curve through 2π radians about the x -axis from $x = 0$ to $x = 1$.

(b) Calculate the exact value of the volume generated.

3

A solid is formed by rotating part of the curve with equation $y = \frac{1}{\sqrt{1+x^2}}$ about the x -axis through 2π radians, from $x = 0$ to $x = a$.

The value of the volume of the solid is $\frac{\pi^2}{3}$.

Determine the value of a .

5

LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

JOIN THE CHANNEL!

GAUSSIAN ELIMINATION

(a) Use Gaussian elimination to express z in terms of λ for the system of equations:

$$\begin{aligned}x - y - 3z &= 1 \\2x - 3y - 5z &= 8 \\x + 2y + \lambda z &= -7.\end{aligned}$$

4

(b) State the value of λ for which this system is inconsistent.

1

(c) Determine the solution of this system when $\lambda = -1$.

1

A system of equations is defined by

$$\begin{aligned}x - 3y + z &= -1 \\3x - 2y + 4z &= 11 \\x + 4y + 2z &= 15\end{aligned}$$

Use Gaussian elimination to determine whether the system shows redundancy, inconsistency or has a unique solution.

3

MATRICES INVERSE

Matrix A is defined by $A = \begin{pmatrix} 6 & 1 \\ 11 & 3 \end{pmatrix}$.

(a) Find A^{-1} , the inverse of matrix A .

2

Matrix B is defined by $B = \begin{pmatrix} -4 & 3 \\ -5 & 2 \end{pmatrix}$.

(b) Find the matrix M such that $AM = B$.

2

LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

JOIN THE CHANNEL!

MATRICES INVERSE

Matrix A is defined by $A = \begin{pmatrix} 2 & 2x & 4 \\ x & -1 & 0 \\ 1 & 0 & -2 \end{pmatrix}$, where $x \in \mathbb{R}$.

- (a) Find a simplified expression for the determinant of A . 2
- (b) Hence, determine whether A^{-1} exists for all values of x . 1

MATRIX OPERATIONS

Matrices A and B are defined by $A = \begin{pmatrix} -3 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2 \\ 5 & \lambda \end{pmatrix}$ where $\lambda \in \mathbb{R}$.

- (a) Find $3A + 2B$. 1
- (b)
 - (i) Find $A'B$, where A' is the transpose of A . 2
 - (ii) Find an expression for the determinant of $A'B$. 1
 - (iii) Determine the value of λ such that $A'B$ is singular. 2

The matrix A has the following property:

$$A^2 = 6A - I, \text{ where } I \text{ is the identity matrix.}$$

- (a) Express A^3 in the form $pA + qI$, where $p, q \in \mathbb{R}$. 2

Matrix A is non-singular.

- (b) Find a similar expression for A^{-1} in terms of A and I . 2

MATRIX TRANSFORMATIONS

- (a) Find the 2×2 matrix, A , associated with a reflection in the x -axis. 1
- (b) Describe the transformation associated with the matrix $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. 1
- (c) Find the 2×2 matrix, C , associated with a reflection in the x -axis followed by the transformation associated with $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. 2

- (a) State the matrix A , associated with an anti-clockwise rotation of $\frac{\pi}{2}$ radians about the origin. 1

The matrix B is given by

$$B = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

The matrix given by AB is associated with an anti-clockwise rotation of α radians about the origin.

- (b) (i) Determine AB . 1
- (ii) Find the value of α . 1
- (c) Determine the least positive integer value of n such that $(AB)^n = I$, where I is the 2×2 identity matrix. 1

COMPLEX NUMBERS

MODULUS AND ARGUMENT

A complex number is defined by $z = 1 + i$.

- (a) Express z in polar form. 2
- (b) Use de Moivre's theorem to evaluate z^8 . 2

- (a) Express $z = 1 + \sqrt{3}i$ in polar form. 2
- (b) Hence, or otherwise, show that z^3 is real. 2

COMPLEX NUMBERS - OPERATIONS

Two complex numbers are defined as $z = 11 + 10i$ and $w = 3 - 2i$.

Find $\frac{z}{w}$ in the form $a + bi$, where $a, b \in \mathbb{R}$. 2

Given that $z_1 = 5 + 3i$ and $z_2 = 6 + 2i$, express $z_1 \bar{z}_2$ in the form $a + ib$ where a and b are real numbers. 2

COMPLEX NUMBERS - SOLVING EQUATIONS

Given $z = x + iy$, $y \neq 0$, solve the equation

$$z^2 + 20\bar{z} - 156 = 0$$

where $\bar{z}$ is the complex conjugate of z . 5

The complex number $z = 3 + i$ is a root of $z^2 - 6z + a = 0$, where a is a real number.

- (a) State the second root of $z^2 - 6z + a = 0$. 1
- (b) Hence, or otherwise, find the value of a . 2

The expression $z^2 - 6z + a$ is a factor of $z^3 - z^2 - 20z + b$, where b is a real number.

- (c) Find the value of b . 1

COMPLEX NUMBERS – SOLVING EQUATIONS

The complex number $z=2+i$ is a root of the polynomial equation $z^4 - 6z^3 + 16z^2 - 22z + q = 0$, where $q \in \mathbb{Z}$.

- (a) State a second root of the equation. 1
- (b) Find the value of q and the remaining roots. 6
- (c) Show the solutions to $z^4 - 6z^3 + 16z^2 - 22z + q = 0$ on an Argand diagram. 1

COMPLEX NUMBERS – LOCUS

Given $z = x + iy$, sketch the locus in the complex plane given by $|z| = |z - 2 + 2i|$.

3

COMPLEX NUMBERS – DE MOIVRE'S THEOREM

Let $z = x + iy$ be a complex number, where $x, y \in \mathbb{R}$.

- (a) (i) Express $\bar{z} + iz$ in Cartesian form, where $\bar{z}$ is the complex conjugate of z . 2
- (ii) Given $x > y$, find the argument of $\bar{z} + iz$. 1

When $x < y$, $\bar{z} + iz = r \left(\cos \left(-\frac{3\pi}{4} \right) + i \sin \left(-\frac{3\pi}{4} \right) \right)$ where r is the modulus of $\bar{z} + iz$.

- (b) Use de Moivre's theorem to find, in polar form, both square roots of $\bar{z} + iz$. 2

LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

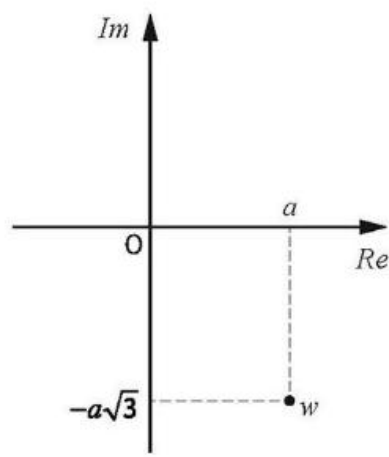
JOIN THE CHANNEL!

COMPLEX NUMBERS – DE MOIVRE’S THEOREM

Let $z = \cos \theta + i \sin \theta$.

- (a) Use de Moivre’s theorem to state an expression for z^4 . 1
- (b) State and simplify the binomial expansion of $(\cos \theta + i \sin \theta)^4$. 3
- (c) Hence show that:
 - (i) $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$. 2
 - (ii) $\sin \theta \cot 4\theta$ can be written in terms of $\cos \theta$ only. 2

The complex number w has been plotted on an Argand diagram, as shown below.



- (a) Express w in
 - (i) Cartesian form 1
 - (ii) polar form. 3

(b) The complex number z_1 is a root of $z^3 = w$, where

$$z_1 = k \left(\cos \frac{\pi}{m} + i \sin \frac{\pi}{m} \right)$$

for integers k and m .

Given that $a = 4$,

- (i) use de Moivre’s theorem to obtain the values of k and m , and 4
- (ii) find the remaining roots. 2

EUCLIDEAN ALGORITHM

(a) Use the Euclidean algorithm to find d , the greatest common divisor of 1118 and 416.	1
(b) Hence find integers a and b such that $1118a + 416b = d$.	2
(a) Use the Euclidean algorithm to find d , the greatest common divisor of 703 and 399.	1
(b) Find integers a and b such that $d = 703a + 399b$.	2
(c) Hence find integers p and q such that $76 = 703p + 399q$.	1

NUMBER BASES

Express 572_{10} in base 9.	2
Express 231_{11} in base 7.	3

PROOF – DIRECT AND COUNTER

Consider statements A and B below.	
For each statement: if true, provide a proof; if false, provide a counterexample.	3
A: The sum of the squares of any two consecutive integers is always prime.	
B: The sum of the squares of any two consecutive integers is always odd.	

(a) Consider the statement:	
For all odd numbers n , $n^2 + 4$ is prime.	
Find a counterexample to show that the statement is false.	1
(b) Prove directly that the difference between the cubes of any two consecutive integers is not divisible by 3.	3

PROOF – CONTRAPOSITIVE

Use the contrapositive to prove that if n^2 is a multiple of 3 then n is a multiple of 3.

Prove by contrapositive that if pq is irrational then at least one of p or q is irrational.

APPLYING SUMMATION FORMULA

Find and fully factorise an expression for $\sum_{r=1}^n (r^3 - 3r)$. 2

(a) Find an expression for $\sum_{r=1}^n (r^2 + 3r)$ in terms of n .
Express your answer in the form $\frac{1}{3}n(n+a)(n+b)$. 2

(b) Hence, or otherwise, find $\sum_{r=11}^{20} (r^2 + 3r)$. 2

PROOF – BY INDUCTION

Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n \frac{1}{(2r+1)(2r-1)} = \frac{n}{2n+1}. \quad 5$$

Prove by induction that, for all positive integers n , $\sum_{r=1}^n 2^{r-1}r = 2^n(n-1) + 1$. 5

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

PROOF – BY INDUCTION

The matrix A is given by $A = \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix}$.

Prove by induction that

$$A^n = \begin{pmatrix} 3^n & 1-3^n \\ 0 & 1 \end{pmatrix}, \forall n \in N.$$

5

BINOMIAL THEOREM – EXPANDING BRACKETS

Use the binomial theorem to expand $\left(\frac{1}{x} - 3x\right)^4$.

Simplify your answer.

4

BINOMIAL THEOREM – GENERAL TERM

(a) State and simplify the general term in the binomial expansion of $\left(2x^2 - \frac{1}{x^3}\right)^{16}$.

3

(b) Hence, or otherwise, find the coefficient of $\frac{1}{x^{18}}$ in the expansion of $\left(2x^2 - \frac{1}{x^3}\right)^{16}$.

2

(a) Write down and simplify the general term in the binomial expansion of

$$\left(3x - \frac{2}{x^2}\right)^8.$$

3

(b) Hence, or otherwise, determine the coefficient of x^{-1} .

2

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

SEQUENCES AND SERIES - ARITHMETIC

An arithmetic sequence has first term -3 and common difference d .

(a) State an expression for the third term. 1

The eighth term is five times the third term.

(b) Find the value of d . 1

(c) Determine algebraically the least number of terms required so that the sum of the associated series is greater than 500. 3

The first three terms of a sequence are defined algebraically by

$$x+5, 3x+2, 5x-1, \text{ where } x \in \mathbb{N}.$$

(a) Show that these three terms form the start of an arithmetic sequence. 2

(b) Find a simplified expression for the 15th term of this sequence. 2

(c) Given that the sum of the first 20 terms of this sequence is 1130, find the value of x . 2

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

SEQUENCES AND SERIES - GEOMETRIC

An infinite geometric sequence of positive numbers has second term 100 and fourth term 16.	
(a) Determine:	
(i) the common ratio	2
(ii) the first term of this sequence.	1
(b) Explain why the associated geometric series has a sum to infinity.	1
(c) Determine this sum to infinity.	1
A new geometric sequence is formed by multiplying each term in the sequence above by the real number k , where $k \neq 0$.	
(d) State the effect that this will have on:	
(i) the common ratio	1
(ii) the sum to infinity of the associated series.	1

A geometric sequence of positive terms has third term 36 and fifth term 16.	
(a) Calculate the value of the common ratio.	2
(b) Calculate the value of the first term.	1
(c) State why the associated geometric series has a sum to infinity.	1
(d) Find the value of this sum to infinity.	1

MACLAURIN SEQUEUNCES

- (a) Find and simplify the Maclaurin expansion, up to and including the term in x^4 , for $\cos 3x$. 2
- (b) Hence find and simplify the Maclaurin expansion, up to and including the term in x^4 , for $\cos^2 3x$. 2

- (a) Find and simplify the Maclaurin expansion, up to and including the term in x^3 , for:
 - (i) e^{2x} 2
 - (ii) $\sin 3x$. 2
- (b) Hence find the Maclaurin expansion for $e^{2\sin 3x}$ up to and including the term in x^3 . 2

A function $f(x)$ has the following properties:

- $f'(x) = \frac{x+1}{1+(x+1)^4}$
- the first term in the Maclaurin expansion of $f(x)$ is 1.

- (a) Find the Maclaurin expansion of $f(x)$ up to and including the term in x^2 . 3
- (b) Use the substitution $u = (x+1)^2$ to find $\int \frac{x+1}{1+(x+1)^4} dx$. 3
- (c) Determine an expression for $f(x)$. 2

FUNCTIONS – ODD AND EVEN

The function $f(x)$ is defined by $f(x) = x^3 - x$, $x \in \mathbb{R}$.

(a) Determine whether $f(x)$ is even, odd or neither. 2

(b) Show that the graph of $y = f(x)$ has a point of inflection. 2

FUNCTIONS – ASYMPTOTES

On a suitable domain a curve is given by the equation $y = f(x)$, where

$$f(x) = \frac{x^2 + x + 5}{x - 2}.$$

(a) Express $f(x)$ in the form $Ax + B + \frac{C}{x - 2}$, where A , B and C are constants. 2

(b) State the equations of the vertical and non-vertical asymptotes of the curve. 2

A function is defined on a suitable domain by $f(x) = \frac{3x^2 + 2}{x^2 - 2}$.

(a) Obtain equations for the asymptotes of the graph of $y = f(x)$. 3

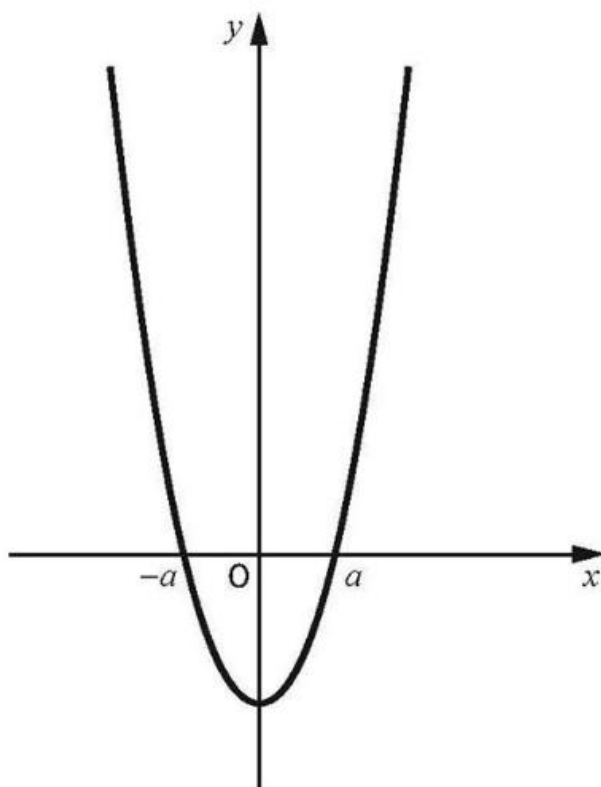
(b) Determine whether the graph of $y = f(x)$ has any points of inflection.
Justify your answer. 5

**LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM**

JOIN THE CHANNEL!

FUNCTIONS – RELATED

The function $f(x)$ is defined by $f(x) = x^2 - a^2$. The graph of $y = f(x)$ is shown in the diagram.



(a) State whether $f(x)$ is odd, even or neither. Give a reason for your answer.

1

(b) Sketch the graph of $y = |f(x)|$.

1

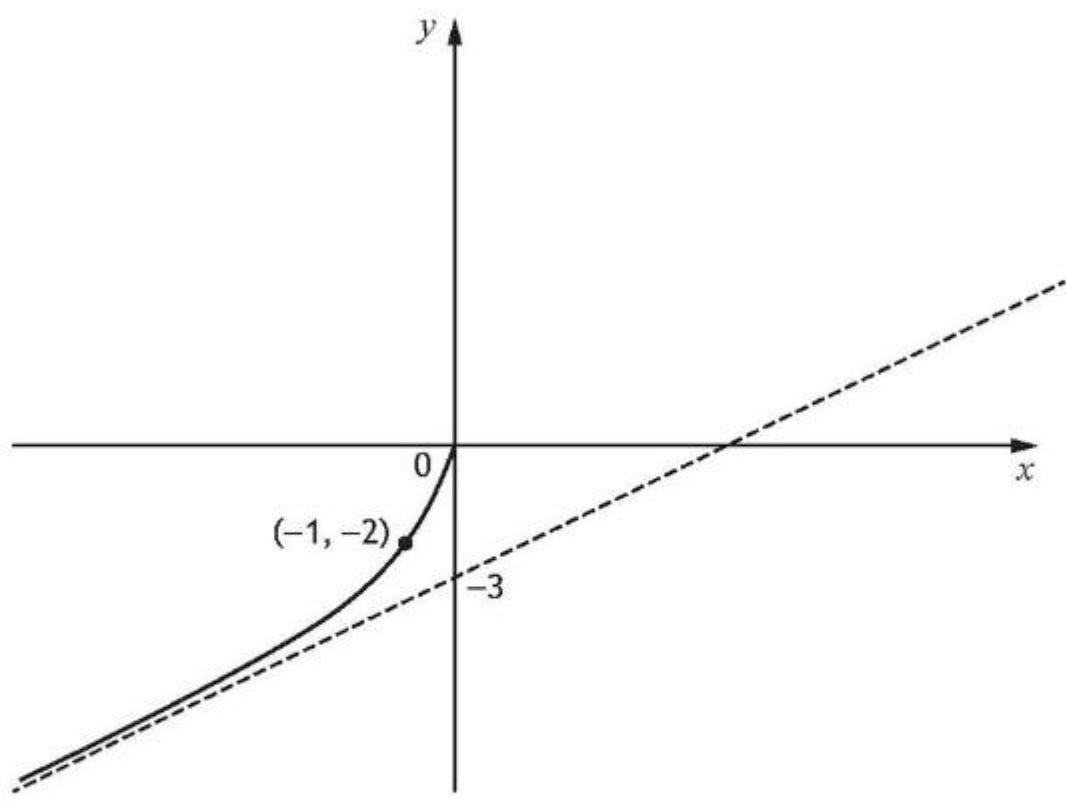
LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

JOIN THE CHANNEL!

FUNCTIONS – RELATED

In the diagram below part of the graph of $y = f(x)$ has been omitted.

The point $(-1, -2)$ lies on the graph and the line $y = \frac{1}{2}x - 3$ is an asymptote.



Given that $f(x)$ is an odd function:

- (a) Copy and complete the diagram, including any asymptotes and any points you know to be on the graph. 2
- (b) $g(x) = |f(x)|$. On a separate diagram, sketch $g(x)$.
Include known asymptotes and points. 2
- (c) State the range of values of $f'(x)$ given that $f'(0) = 2$. 1

VECTORS – EQUATIONS OF LINES OR PLANES

Three planes are defined by

$$\pi_1: x - y + 4z = -6$$

$$\pi_2: 2x + 3y + z = 15$$

$$\pi_3: 3x + 2y - 2z = 16$$

(a) Use Gaussian elimination to find T, the point of intersection of the three planes. 4

The line L_1 is defined by $\frac{x+4}{3} = \frac{y-7}{2} = \frac{z-4}{1}$.

(b) Find P, the point of intersection of the line L_1 and the plane π_3 . 3

The line L_2 passes through points T and P.

(c) Find, in parametric form, the equations of the line L_2 . 2

A plane passes through A (2, -1, 8), B (1, 1, -1) and C (4, -2, 11).

(a) (i) Determine $\overrightarrow{AB}$ and $\overrightarrow{AC}$. 1

(ii) Hence find the Cartesian equation of the plane. 3

A line is defined by the equations $\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z+1}{4}$.

(b) Show that the line and the plane do not intersect. 3

VECTORS – INTERSECTIONS OF LINES

The equations of two planes are given below.

$$\pi_1: 2x - 3y - z = 9$$

$$\pi_2: x + y - 3z = 2$$

- (a) Verify that the line of intersection, L_1 , of these two planes has parametric equations

$$x = 2\lambda + 3$$

$$y = \lambda - 1$$

$$z = \lambda$$

2

- (b) Let π_3 be the plane with equation $-2x + 4y + 3z = 4$.
Calculate the acute angle between the line L_1 and the plane π_3 .

3

- (c) L_2 is the line perpendicular to π_3 passing through $P(1, 3, -2)$.
Determine whether or not L_1 and L_2 intersect.

4

VECTORS – INTERSECTIONS OF PLANES

Planes π_1 , π_2 and π_3 have equations:

$$\pi_1: x - 2y + z = -4$$

$$\pi_2: 3x - 5y - 2z = 1$$

$$\pi_3: -7x + 11y + az = -11$$

where $a \in \mathbb{R}$.

- (a) Use Gaussian elimination to find the value of a such that the intersection of the planes π_1 , π_2 and π_3 is a line.
- (b) Find the equation of the line of intersection of the planes when a takes this value.

4

2

The plane π_4 has equation $-9x + 15y + 6z = 20$.

- (c) Find the acute angle between π_1 and π_4 .
- (d) Describe the geometrical relationship between π_2 and π_4 .
Justify your answer.

3

1



JOIN NOW

DONE THE PRACTICE? NOW LET'S LOCK IN THE GRADE.



AH 5 MATHS LAST MINUTE LIVE STREAM.

What You Get

- 🧠 Mr. Clelland's 100% Original Predicted Questions.
- 💬 Live, real-time Q&A to solve your exact problems.
- ⌚ Final exam technique and time-management hacks.

Note: The Live Stream is an independent Clelland Maths event and uses entirely original material. It does not contain SQA copyrighted questions.

LAST MINUTE ADVANCED HIGHER MATHS LIVE STREAM
WEDNESDAY 6TH MAY 7:30PM

JOIN THE CHANNEL!